

PROFILE

Mechanical Engineering graduate (BEng, First Class Honours) with hands-on experience across CAD design, embedded systems, and additive manufacturing, taking projects from concept through iterative prototyping to functional, tested hardware. Seeking an entry-level engineering role applying rigorous design and analytical skills to real-world engineering challenges.

KEY SKILLS

- **CAD & Product Design:** SolidWorks, Fusion 360, iterative design, design for manufacture
- **Embedded Systems & Electronics:** Arduino, Raspberry Pi, GPIO programming, sensor integration and calibration, circuit design, analogue electronics
- **Programming & Automation:** Python, PLC programming (Siemens TIA Portal), data logging, generative AI tools for workflow efficiency
- **Machine Learning & Computer Vision:** YOLOv8 object detection, model training and evaluation
- **Manufacturing & Prototyping:** Additive manufacturing / 3D printing, rapid prototyping, functional testing
- **Engineering Analysis:** Stress analysis and design, control systems
- **Other:** Technical writing and reporting, MS Office

EDUCATION & QUALIFICATIONS

University of East Anglia — BEng Mechanical Engineering (Hons) 2023 – 2026

- First Class Honours (76.3%)
- IMechE Best Student Certificate 2025 – awarded for top academic performance, Year 2
- Key modules: Control Systems, Analogue and Digital Electronics, Stress Analysis and Design, Mechanical Concepts and Applications

City College Norwich — Access to Higher Education Diploma, Engineering 2022 – 2023

- Distinction (60 Level 3 credits)

City College Norwich — A Levels: Biology, Chemistry 2019 – 2021

PROJECTS

Final Year Project – Wearable Sensor Footwear System 2025 – 2026

- Engineered a 3D-printed wearable footwear system, applying iterative design, CAD modelling, and additive manufacturing to refine form, function, and sensor placement.
- Integrated embedded sensors and electronics within the mechanical structure, covering the full product development cycle from concept to validated device.
- Delivered a fully functional, tested prototype, achieving a 77% project report grade.

Smart Pillbox System – Embedded Systems & Computer Vision Project 2025 – 2026

- Co-developed a Raspberry Pi-based smart pillbox proof of concept for medication adherence monitoring, combining camera-based pill detection, load cell weight sensing, and physical LED/buzzer feedback.
- Designed and calibrated a load cell and HX711 amplifier circuit for weight-based detection, and built a lid-latch mechanism in Fusion 360 to trigger sensor readings automatically on closure.
- Trained and deployed a YOLOv8 object detection model for pill counting, achieving 90% precision and 83.6% recall on a 2,000+ image dataset, exported the model to NCNN format for real-time inference on the Raspberry Pi, and earned an 83% project grade.

Analogue Electronics Project – AM Communication System 2024 – 2025

- Built and tested an analogue AM communication system to transmit and receive audio signals across a metal conductor, achieving reliable, validated signal transmission.
- Implemented core analogue electronics, including 555 timer-based oscillators and op-amp summing amplifiers.

Personal Projects

- Leveraged CAD design and rapid prototyping to create practical 3D-printed household solutions.
- Developed small electronic circuits and microcontroller-based systems through independent DIY projects.

EXPERIENCE

Formula Student – Electronics & Controls Team Dec 2025 – May 2026

- Established early-stage sensor selection and integration strategy as a founding member of the university's Formula Student Electronics and Controls team.
- Applied embedded systems and control systems knowledge to foundational project planning within a newly formed multidisciplinary engineering team.

Customer Advisor | B&Q Nov 2021 – May 2023

- Awarded Employee of the Month within 6 months of joining, recognised for customer communication and teamwork.
- Provided efficient, accurate customer advice and timber-cutting service within the building department.
- Collaborated within a small team to manage orders, maintain departmental standards, and support customers efficiently during peak trading hours.